



Diploma in Software Engineering 26.1F
Module: DATA MANAGEMENT 1

PROJECT PROPOSAL

DATA MANAGEMENT 1
COURSE WORK

B2B TEXTILE INVENTORY AND SALES MANAGEMENT SYSTEM
Organization: Colour Selection – Textile Trading and Reselling

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1. Introduction & Organization Background

Running a growing business-to-business textile operation means handling different parts every day. It starts with getting materials like cotton, silk and special threads from suppliers that are both nearby and far away. Then there is the job of making sure regional warehouses have stock for local clothing makers and fashion stores. Colour Selection's everyday work depends , on keeping this chain of supply working well and making sure all orders are handled correctly.

2. Problem Statement

Like growing trading businesses Colour Selection used to manage inventory, sales and supplier details with a patchwork of disconnected spreadsheets and manual records. This naturally caused a major hurdles, for our team:

- **Inventory Blind Spots:** Inability to monitor real-time stock levels across different warehouses, leading to potential stockouts or overselling.
- **Disconnected Financial Records:** Sales orders, billing invoices, and payment tracking are handled separately, making it difficult to track unpaid client invoices or pending order statuses.
- **Supplier & Product Mismanagement:** Difficulty in quickly associating specific products with their corresponding suppliers and managing unit pricing effectively.

3. Project Scope and Objectives

The goal of this project is to design, develop and build a normalized Relational Database Management System using Microsoft SQL Server (SSMS). The system will automate operational tasks by creating a strong secure database. This database will include nine normalized entities that form the core of the system.

1. **Supply Chain Flow:** Tracking **Suppliers, Products, Warehouses**, and **Inventory** stock levels.
2. **Sales & Billing Flow:** Tracking **Customers, Sales Orders, Order Details, Invoices**, and **Payments**.

4. System Assumptions & Out of Scope

- **Automated B2B Portal:** The system operates under the assumption of a modern, automated B2B transaction environment where customer orders and billing are processed centrally through system workflows.
- **Exclusion of HR / Employees:** To maintain a strict focus on supply chain efficiency and financial tracking, Human Resources and individual employee management entities are excluded from the database scope.

5. Expected Outcomes & Security Design

- **Data Integrity:** Complete elimination of data redundancy through 3rd Normal Form (3NF) relational mapping, enforced via Primary and Foreign Key constraints.
- **Business Intelligence:** Implementation of 13 complex SQL queries (including multi-table JOINS and aggregate functions with GROUP BY, ORDER BY, and HAVING) to generate instant reports on stock, sales revenue, and unpaid balances.
- **Role-Based Security:** Implementation of database-level security through distinct user roles (TextileManager and TextileClerk) with granular GRANT and DENY permissions to protect sensitive financial records.